Submission Summary

Conference Name

International Conference on Recent Advances in Multidisciplinary Engineering 2026

Track Name

Computer Science & Engineering

Paper ID

169

Paper Title

TrustTag: A Lost Item Finder App Using QR Code and Location Tracking

Abstract

—The loss of personal valuables like mobile phones, wallets, keys and cards is a common and growing problem worldwide. Research indicates that over 30 million cell phones are lost each year, and less than 25% recovered, highlighting the importance of an effective technology-based recovery system. Traditional retrieval techniques (e.g., notice boards, social media posting and manual search) are not only unreliable and time consuming, but also significantly reveal the users' privacy. Current technological options like RFID technologies [1], Bluetooth Low Energy (BLE) trackers [3]–[4], and IoT–GPS-based devices have hardware dependence, ecosystem constraints, infrastructure costs, and privacy issues [7] that prevents being adopted at a general level. This paper introduces the Lost Item Finder App; an Android application-based solution that offers a cost-effective, hardware-less approach for dynamic QR codes, including GPS location capture support and multi-channel real-time alerts in order to quickly and safely recover lost items. Personal items are registered and tagged with QR code, which each user can customize by cryptographic tokenization. Unlike the static QR tags as adopted in library and healthcare systems [9], [10], we use connected cloud driven dynamic QR codes in the proposed system, which are activated when an item is reported lost to maintain moderate information disclosure. When scanned, finders can see some user-approved contact information and owners are immediately alerted via in-app notifications, SMS text and with addition to GPS coordinates. The backend is implemented in Firebase with Flask, using HTTPS and AES 256 encryption to prevent privacy and security issues raised by BLE and IoT research [4], [6], [7].

Index Terms—Lost Item Finder, QR Code, GPS, Android, SOS Alert, Privacy, Real-Time Notifications.

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Primary Subject Area

AI, ML, Cloud Computing, IoT, Blockchain, Cybersecurity, Big Data, Computer Vision, Software Engineering.

Submission Files

Research Paper.pdf (211.5 Kb, 2/13/2026, 7:03:31 AM)

Submission Questions Response

1. Keywords (Required)

List 3–6 keywords relevant to your paper.

Lost Item Recovery, OR Code Technology, Real-Time Location Tracking, Android application, GPS-Based Safety System

2. Ethical Compliance (Required)

Confirm that your research adheres to ethical standards (human/animal data, privacy, consent).

Agreement accepted

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